

APPENDIX B—NOMENCLATURE

B.0 General

This Appendix describes the nomenclature used in these Regulations in three parts. Subsection B.1 describes the Units of Measurement. Subsection B.2 lists the Symbols, Abbreviations and Cargo IMP Codes. Subsection B.3 contains tables of useful conversion factors to and from SI units.

B.1 Units of Measurement

B.1.1 Use of SI Units

The units of measurement to be used in the transport of dangerous goods by air are those specified by the International System (SI) as modified for international civil aviation by *Annex 5 to the Chicago Convention on International Civil Aviation*. The primary units of length, weight and volume will be the metre (m), the kilogram (kg) and the litre (L) and the unit of pressure will be the kilopascal (kPa). Where measurements relating to radioactivity occur in these Regulations, the values are shown in SI units followed by the non-SI equivalents in parentheses. The units used must always be expressed together with the quantities. Except as specifically provided for in these Regulations, only those abbreviations for units of measurement that are indicated in this section may be used in the transport of dangerous goods by air.

Note:

For small quantities, decimals of the above units, for example centimetre (cm) or millimetre (mm), gram (g) or milligram (mg) etc. may be used provided the units are always clearly specified.

B.1.2 Non-SI Equivalent

It is recognized that there are in existence many packagings which were designed and constructed for use with non-SI quantity limitations and that such packagings will continue to be used for some time to come. Table B.3.C therefore contains a list of authorized non-SI equivalents for quantity limitations expressed in SI units. It is stressed that these are not precise equivalents but are nevertheless acceptable based upon the likely availability of packaging.

Note:

All information relating to dangerous goods quantities must be provided in SI units.

B.1.3 Conversion Factors and Authorized Equivalents

Precise conversion factors for commonly used SI units are given in *Annex 5 to the Chicago Convention*. Tables B.3.A and B.3.B show conversion factors, to four

significant figures, for some units widely used in dangerous goods transport.

B.1.4 Number Format

The preferred decimal marker is a point on the line (period), e.g. 2.5 kg. However, use of the European format with a comma is also acceptable, e.g. 2,5 kg. When writing numbers less than one, a zero should be written before the decimal marker.

B.1.5 Pressure

Pressures of all kinds related to cylinders (such as test pressure, internal pressure) are always indicated in gauge pressure (pressure in excess of atmospheric pressure); however, the vapour pressure of substances is always expressed in absolute pressure.

B.2 Symbols and Abbreviations

B.2.1 Symbols

The following symbols have the meanings shown:

Symbol—Meaning

†—Indicates brief description in Appendix A. (Note: this symbol is not part of the proper shipping name).

★—The technical name or chemical group name(s) is required to be shown, in parentheses, following the proper shipping name. (Note: this symbol is not part of the proper shipping name).

⚡—Indicates a more restrictive requirement than ICAO

Y—Prefix letter indicating a Limited Quantity Packing Instruction. Also used to indicate compliance with the air transport requirements in the limited quantities mark.

>—Greater than

≥—Equal to or greater than

<—Less than

≤—Equal to or less than

□—Indicates addition of new item

△—Indicates change in this item

⊗—Indicates cancellation of item previously shown in this space

☢—Indicates that the item relates entirely to Radioactive Shipments. Used in Subsection 2.8, State and operator variations and Appendix A—Glossary.

B.2.2 Abbreviations

B.2.2.1 General Abbreviations

The following general abbreviations have the meanings shown:

Abbreviation—General Term

CSI—Criticality Safety Index
 Dang.—Dangerous (List 4.2, Column D)
 Desc.—Description
 Div.—Division
 e.g.—for example
 EQ—Excepted Quantity (List 4.2, Column F)
 ERG—Emergency Response Guide (List 4.2 Column N)
 etc.—and so forth
 Fl.—fluid
 Flamm.—Flammable (List 4.2, Column D)
 G—Gross mass or weight of package as prepared for transport
 IBC—Intermediate Bulk Container
 ID—Identification number
 i.e.—that is
 Imp.—Imperial
 IP—inner packaging
 ISO—International Organization for Standardization
 LC—lethal concentration
 LD—lethal dose
 LPG—liquefied petroleum gas
 LTD QTY—Limited Quantity of Dangerous Goods (see 2.7)
 max—maximum
 N/A—not applicable
 NEC—Net Explosive Content
 NEM—Net Explosive Mass
 NEQ—Net Explosive Quantity
 NEW—Net Explosive Weight
 n.o.s.—not otherwise specified
 PG—Packing Group
 PI—Packing Instruction
 Pkg—Packing
 Pkg Instr.—Packing Instruction
 Pkge—Package
 ppm—parts per million (i.e. mg/kg)
 PPR—Particular Packing Requirement
 Qty—Quantity

SADT—the lowest temperature at which self accelerating decomposition may occur with a substance in the packaging as used in transport

SI—International System of Units

S.P.—Special Provision (List 4.2, Column M)

Spec.—Specification

Spont. Comb.—Spontaneously Combustible (List 4.2, Column D)

TI—Transport Index

UN—UN number assigned by the United Nations Committee of Experts on the Transport of Dangerous Goods.

B.2.2.2 Technical Abbreviations

The following abbreviations for technical terms have the meanings shown:

Abbreviation—Technical Term

A/m—amperes per metre
 bar—bar = unit of pressure in liquids/gases
 Bq—becquerel = unit of radioactivity
 °C—degree Celsius
 Ci—curie = unit of radioactivity
 cm—centimetre
 cm²—square centimetre
 ft—feet
 g—gram
 g/m²—grams per square metre
 Gy—gray
 h—hour
 Hz—hertz
 in—inch
 J/kg—joule per kilogram
 K—Kelvin = unit of temperature
 keV—kilo electron volt
 kg—kilogram
 kgf—kilogram force
 kg/m²—kilogram per square metre
 kPa—kilopascal
 L—litre
 L—List 4.2 column N only, ERG (q.v.) code letter for low/other hazard
 m—metre
 m²—square metre
 m³—cubic metre
 µm—micrometre
 MeV—mega electron volt
 mCi—millicurie

nCi—nanocurie
 µCi—microcurie
 mg—milligram
 mL—millilitre
 mm—millimetre
 mm²/s—square millimetre per second = unit of kinematic viscosity
 mR/h—milliroentgen per hour
 mrem/h—millirem per hour
 mS/m—millisiemens per metre
 mSv—millisievert
 µSv/h—microsieverts per hour
 N—newton = unit of force
 N/mm²—newton per square millimetre = unit of mechanical strength (tensile, compressive strength)
 Ω /m—ohm per metre
 Pa—pascal = unit of pressure or stress
 R/h—roentgen per hour
 rem/h—rem per hour
 s—second = unit of time
 Sv—sievert
 Wh—Watt-hour
 W/m²—watts per square metre
 W/m/K—watts per metre per Kelvin

B.2.2.3 Prefixes for Decimal Multiples of SI Units

The following prefixes and symbols are used for decimal multiples of SI Units:

Prefix	Symbol	Multiplication Factor
peta	P	1 000 000 000 000 000 = 10 ¹⁵
tera	T	1 000 000 000 000 = 10 ¹²
giga	G	1 000 000 000 = 10 ⁹
mega	M	1 000 000 = 10 ⁶
kilo	k	1 000 = 10 ³
hecto	h	100 = 10 ²
deka	da	10 = 10 ¹
deci	d	0.1 = 10 ⁻¹
centi	c	0.01 = 10 ⁻²
milli	m	0.001 = 10 ⁻³
micro	µ	0.000 001 = 10 ⁻⁶
nano	n	0.000 000 001 = 10 ⁻⁹

B.2.2.4 IATA Cargo IMP Codes (Special Handling Codes)

The following Cargo-IMP Codes (Special Handling Codes) are used extensively within the airline industry and have the meanings shown:

Code—Meaning

CAO—Cargo Aircraft Only
 DGD—Shipper's Declaration for Dangerous Goods
 ELI—Lithium ion batteries excepted as per Section II of PI 966 and 967
 ELM—Lithium metal batteries excepted as per Section II of PI 969 and 970
 ICE—Carbon dioxide, solid (dry ice)
 IMP—Interchange Message Procedure
 MAG—Magnetized Material
 RBI—Fully regulated lithium ion batteries (Class 9, UN 3480) as per Section IA and IB of PI 965, and where applicable lithium ion batteries shipped under an approval in accordance with special provision A88 or A99
 RBM—Fully regulated lithium metal batteries (Class 9, UN 3090) as per Section IA and IB of PI 968, and where applicable lithium metal batteries shipped under an approval in accordance with special provision A88 or A99
 RCL—Cryogenic Liquid (Packing Instruction 202)
 RCM—Corrosive
 RCX—Explosives 1.3C
 RDS—Biological Substance, Category B (UN 3373)
 REQ—Dangerous Goods in Excepted Quantities
 REX—To be reserved for normally forbidden Explosives, Divisions 1.1, 1.2, 1.3, 1.4F, 1.5 and 1.6
 RFG—Flammable Gas
 RFL—Flammable Liquid
 RFS—Flammable Solid
 RFW—Dangerous When Wet
 RGX—Explosives 1.3G
 RIS—Infectious Substance (UN 2814 or UN 2900)
 RLI—Fully regulated lithium ion batteries (Class 9, UN 3481) as per Section I of PI 966 and 967, and where applicable lithium ion batteries shipped under an approval in accordance with special provision A88 or A99
 RLM—Fully regulated lithium metal batteries (Class 9, UN 3091) as per Section I of PI 969 and 970, and where applicable lithium metal batteries shipped under an approval in accordance with special provision A88 or A99
 RMD—Miscellaneous Dangerous Goods
 RNG—Non-Flammable Non-toxic Gas
 ROP—Organic Peroxide
 ROX—Oxidizer
 RPB—Toxic substance

RPG—Toxic Gas
 RRE—Excepted Packages of Radioactive Material
 RRW—Radioactive Material Category I-White
 RRY—Radioactive Material Categories II-Yellow and III-Yellow
 RSB—Polymeric Beads/Plastics Moulding Compound (Packing Instruction 957)
 RSC—Spontaneously Combustible
 RXB—Explosives 1.4B
 RXC—Explosives 1.4C
 RXD—Explosives 1.4D

RXE—Explosives 1.4E
 RXG—Explosives 1.4G
 RXS—Explosives 1.4S

B.3 Conversion Factors

Precise conversion factors for commonly used SI units are given in *Annex 5 to the Chicago Convention on International Civil Aviation*. Tables B.3.A and B.3.B show conversion factors, to four significant figures, for some units widely used in dangerous goods transport.

TABLE B.3.A
Conversion to SI Units

To convert	to	Multiply by
bar	kilopascal (kPa)	100.0
curie (Ci)	gigabecquerel (GBq)	37.00
temperature in degree Fahrenheit (°F)	temperature in degree Celsius (°C)	subtract 32°F and multiply by 5/9
temperature in degree Celsius (°C)	temperature in Kelvin	add 273.15
foot (ft)	metre (m)	0.3048
gallon (Imperial)	litre (L)	4.546
gallon (US liquid)	litre (L)	3.785
inch (in)	millimetre (mm)	25.40
kilogram per square centimetre (kg/cm ²)	kilopascal (kPa)	98.07
oersted	ampere per meter (A/m)	79.58
ounce (avoirdupois)	gram (g)	28.35
ounce, fluid (Imperial)	millilitre (mL)	28.41
ounce, fluid (US)	millilitre (mL)	29.57
pint (Imperial)	litre (L)	0.5683
pint (US)	litre (L)	0.4732
pound (avoirdupois)	kilogram (kg)	0.4536
pound per square inch (lb/sq.in)	kilopascal (kPa)	6.895
quart (Imperial)	litre (L)	1.137
quart (US)	litre (L)	0.9464
rad	gray (Gy)	0.01000
rem	sievert (Sv)	0.01000

Note:

Atmospheric pressure is the pressure exerted by the atmosphere. Gauge pressure is the pressure of a system measured by a gauge, which excludes atmospheric pressure. Absolute pressure is the sum of the available atmospheric pressure and the gauge pressure of the system.

Absolute Pressure = Gauge Pressure + Atmospheric Pressure.

= 150 psig (gauge pressure) + 14.7 psi (atmospheric pressure) = 164.7 psia, for example.

TABLE B.3.B
Conversion from SI Units

To convert	to	Multiply by
ampere per metre (A/m)	Oersted	0.01257
temperature in degree Celsius (°C)	temperature in degree Fahrenheit (°F)	multiply by 9/5 and add 32°F
temperature in Kelvin	temperature in degree Celsius (°C)	subtract 273.15
gram (g)	ounce (avoirdupois)	0.03527
gray (Gy)	rad	100.00
kilogram (kg)	pound (lb)	2.205
kilopascal (kPa)	bar	0.01000
kilopascal (kPa)	kilogram per square centimetre (kg/cm ²)	0.01020
kilopascal (kPa)	pound per square inch	0.1450
litre (L)	gallon (Imperial)	0.2200
litre (L)	gallon (US liquid)	0.2642
litre (L)	pint (Imperial)	1.760
litre (L)	pint (US)	2.113
litre (L)	quart (Imperial)	0.8799
litre (L)	quart (US)	1.057
metre (m)	foot (ft)	3.281
millilitre (mL)	ounce, fluid (Imperial)	0.03520
millilitre (mL)	ounce, fluid (US)	0.03381
millimetre (mm)	inch (in)	0.03937
sievert (Sv)	rem	100.00
terabecquerel (TBq)	curie (Ci)	27.03

TABLE B.3.C
Authorized Equivalents

Volume		
Litres	Imperial	U.S.
0.5	1 pt	1 pt
1	1 qt	1 qt
2	2 qt	2 qt
2.5	5 pt	5 pt
5	1 gal	1.25 gal
10	2 gal	2.5 gal
15	3 gal	3.75 gal
20	4.25 gal	5 gal
25	5.5 gal	6.25 gal
30	6.5 gal	7.5 gal
42	9 gal	11 gal
50	11 gal	13 gal
60	13 gal	15 gal
100	22 gal	25 gal
120	26 gal	30 gal
220	48 gal	55 gal
250	55 gal	62.5 gal

Note:

Where quantities are specified in SI units of weight, for 500 kg or less, quantities expressed in pounds may be substituted on the basis of 1 kg = 2 lb.

B.4 Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

This appendix provides information on the United Nations Globally Harmonized System of Classification and Labelling of Chemicals (GHS) and provides examples of the pictograms used by the GHS that may appear on the outside of single packagings.

"Globally Harmonized System of Classification and Labelling of Chemicals (GHS)", addresses classification of chemicals by types of hazard and proposes harmonized hazard communication elements, including labels and safety data sheets. It aims at ensuring that information on physical hazards and toxicity from chemicals is available in order to enhance the protection of human health and the environment during the handling, transport and use of these chemicals. The GHS also provides a basis for harmonization of rules and regulations on chemicals at national, regional and worldwide level, an important factor also for trade facilitation.

The two major elements of GHS are classification of the hazards of chemicals according to the GHS rules and communication of the hazards and precautionary information using safety data sheets (SDS) (also known as

material safety data sheets [MSDS]) and labels. The GHS labels may also appear on packages to communicate the hazards in the package by the use of standardized hazard statements, signal words and diamond-shaped pictograms according to the classification of that chemical or mixture. Precautionary statements may also be required.

While some pictograms identify substances that only pose a hazard for supply and use, other GHS pictograms contain symbols that are largely equivalent to the symbols contained in the hazard labels used in transport and which may therefore indicate that the contents are dangerous goods.

The pictograms used by GHS are shown in Tables B.4.A and B.4.B. Table B.4.A identifies the GHS pictograms that contain symbols that also appear on the hazard label used in transport. The presence of these pictograms may indicate that the contents of the package are dangerous goods. Table B.4.B identifies GHS pictograms that identify substances that pose a hazard other than an acute hazard and which do not pose a risk in transport (see also DGR 2.2).

For more information, see http://www.unece.org/trans/danger/publi/ghs/ghs_welcome_e.html.

TABLE B.4.A
GHS Pictograms and their Criteria (B.4)

Pictogram							
Pictogram Name	Explosive	Gases Under Pressure	Flammable	Oxidizer	Toxic	Corrosive	Aquatic Toxicity
Appears on	Explosives		Gases Aerosols Liquids Solids	Oxidizing Gases Oxidizing Liquids Oxidizing Solids	Acute Toxicity Skin Oral Inhalation	Corrosive to Metal Skin Corrosion Serious Eye Damage	Acute Chronic
	Self Reactive Substances and Mixtures		Self Reactive Substances and Mixtures Pyrophoric Liquids and Solids				
	Organic Peroxides		Self-Heating Substances and Mixtures				
			Substances and Mixtures, Which in Contact With Water, Emit Flammable Gases				
			Organic Peroxides				

Notes:

1. The GHS pictograms, as shown above may indicate products which are dangerous goods for transport. There are however national and regional differences which may mean that packages bearing such pictograms are not classified as dangerous goods in transport.
2. When a substance is required to show corrosive symbol based solely on classification criteria of "serious eye damage" in the GHS classification, the substance does not meet the classification criteria of corrosive materials for transport.

TABLE B.4.B
GHS Pictograms and their Criteria (B.4)

Pictogram		
Pictogram Name	Harmful	Respiratory
Appears on	Harmful to Skin Oral Inhalation	Respiratory Sensitization Carcinogenicity Toxic to Reproduction
	Skin Irritation Eye Irritation	Specific Target Organ Systemic Toxicity Single Exposure
	Respiratory Tract Irritation Narcotic Effects	Specific Target Organ Systemic Toxicity Repeated Exposure
	Skin Sensitization	Aspiration Hazard Germ Cell Mutagenicity

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B.4